2024 International Conference on Circuit Power and Computing Technologies (ICCPCT)

Innovative Grid Tied EV Charging using Hybrid Energy Source with Vehicle to Grid and Grid to Vehicle Operation

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2024 7th International Conference on Circuit Power and Computing Technologies (ICCPCT) | 979-8-3503-7281-6/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICCPCT61902.2024.10672736

Abstract—Global power shortages are forcing people to look for other sources of energy to combat the impending power crisis. One potential solution for this is to use grid-synchronized electric vehicles (EVs). When several EVs are linked to the grid, disparities in supply and demand, voltage, and frequency result. This impacts the electricity grid as a whole. To alleviate these concerns, we have combined wind and sun power. V2G technology is one of the advancements in smart grids that allows energy exchange between the electric car and the grid. This proposal proposes using a hybrid energy source to charge EVs. Here, power from the wind turbine, PV panel, and AC generator is used to charge the EV battery. The following household's linear and non-linear needs are subsequently met with any excess electricity. The photovoltaic (PV) panel's DC voltage is maximized using a proportional-integral (PI) controller, which is well-known for its reliability and independence from the real system model. The enhanced DC output of the Boost converter raises the DC voltage overall. In order to provide the grid with additional power, the system also includes an AC generator. The electricity from the PV panel is converted into AC power using a three-phase Voltage Source Inverter (VSI), which uses the enhanced output of the Boost conversion process. This AC electricity is sent into the grid after careful synchronization. An LC filter is used to rectify harmonic components found in the VSI output, ensuring grid compatibility and safe current injection. In this paper, the voltage for the EV battery is adjusted using a bidirectional converter. Electricity from EVs that are not in use sent back into the grid for further applications.

Keywords— EV, V2G, PI Controller, PV System and VSI.

I. INTRODUCTION

By integrating hybrid energy sources with grid-to-vehicle (G2V) and V2G operations, innovative grid-tied EV charging systems offer a significant development in sustainable mobility and energy integration [1, 2]. These innovative solutions use renewable energy and bidirectional energy flow to improve grid stability and resilience while meeting the rising demand for environmentally friendly mobility [3, 4]. Solar panels and wind turbines are sustainable power sources for EV charging since they harvest abundant, renewable energy from the environment. Ultra-capacitors and lithium-ion batteries are examples of energy storage devices that store extra energy for later use when renewable energy generation is insufficient or during peak charging periods. The combination of V2G and G2V capabilities further boosts the systems' adaptability and efficiency [5, 6]. When there are spikes in demand or in reaction to grid signals, EV batteries may use V2G to restore stored energy to the grid [7]. This helps to stabilize the grid and provide ancillary services, all the while earning money for EV owners [8]. By allowing EVs to charge from the grid at off-peak times or periods of high renewable energy output, on the other hand, G2V maximizes energy efficiency and lessens the load on the grid [9, 10].

Innovative grid-tied EV charging systems provide several advantages by fusing renewable energy sources with bidirectional energy flow [11, 12]. Additionally, by offering distributed energy supplies and demand response capabilities, they improve grid resilience and stability. The capabilities of V2G enable EV owners to profit from their car batteries, promote the integration of intermittent renewable energy sources, and aid in grid balancing [13, 14]. The implementation of these technologies signals a revolutionary change in the direction of integrated renewable energy sources and sustainable mobility. They not only tackle the problems of air pollution and climate change, but they also open doors for income production, grid optimization, and the democratization of energy [15-17].

The Contribution of the papers are as follows;

- G2V and V2G power transmission by implementing bidirectional power flow.
- Use excess electricity to support both linear and non-linear loads to meet the energy needs of surrounding families.
- Increase grid resilience by managing excess power and facilitating a bidirectional power flow interface with EVs.

II. PROPOSED SYSTEM DESCRIPTION

This unique solution uses hybrid energy sources to connect the car to the grid for grid-tied electric vehicle charging. A Boost converter is used to inject the ideal DC voltage that retrieved from the PV panel using a PI controller, helping to improvise the DC voltage. Through a (PWM) rectifier, a wind energy conversion system based on a DFIG provides the necessary DC power. Power output from the Boost converter and the improved output from the PWM rectifier are kept in a DC connection and converted to AC via a three-phase VSI before being properly synchronized and fed into the grid.

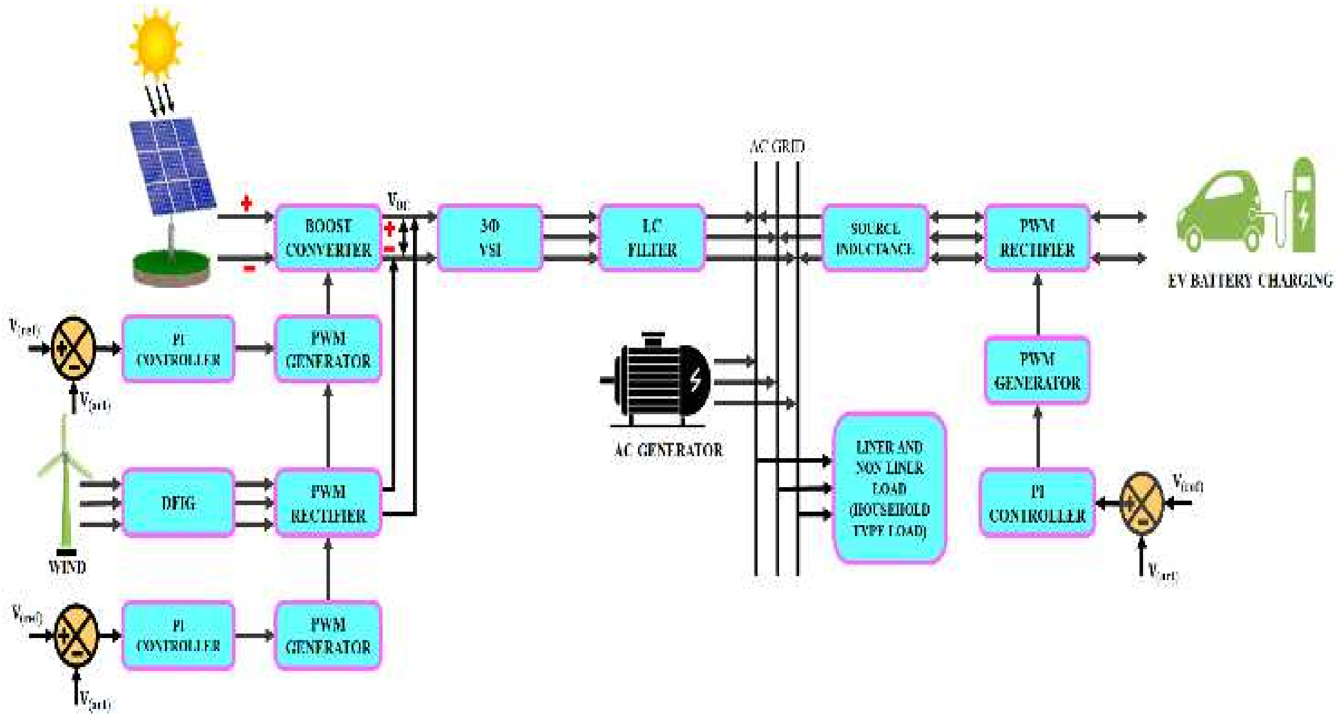

Fig. 1. Proposed Block Diagram

An electric vehicle's stored energy is effectively recycled back into the grid when it is not in use. The batteries of EVs are charged using electricity produced by photovoltaic cells, wind turbines, and AC generators. The excess energy is sent back into the grid for further applications, demonstrating two-way power transfer: G2V and V2G.

III. Proposed System Modelling

A. PV system

A PV system uses solar panels to transform sunlight into electricity, making it a renewable energy source. When exposed to sunlight, semiconductor materials, most often silicon, in solar panels produce direct current (DC) power. Grid-tied PV systems use an inverter to feed the generated power into the electrical grid, which balances or enhances grid-supplied electricity. Off-grid photovoltaic systems, which are popular in isolated locations, store extra power in batteries for future use.

The following is a prediction for the PV solar system's V_i distinctive equation: The PV method's photo-current comes from,

$$I_{pc} = [I_{si} + K_i\,(T - T_o)] * \frac{G}{1000} \tag{1}$$

The present production of the PV component is assumed by

$$I = N_p * I_{pc} - N_p * I_s * \left[exp\left(\frac{\frac{V}{N_S} + I * \frac{R_S}{N_P}}{n * V_{th}}\right) - 1\right] - I_p \tag{2}$$

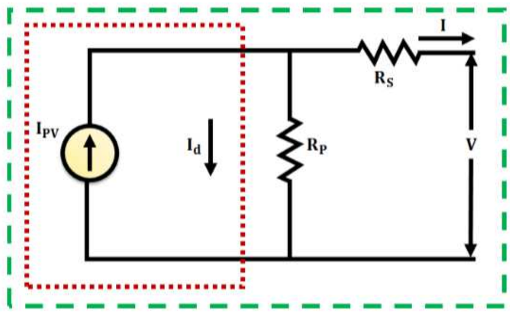

Fig. 2. Circuit Diagram of PV System

Decentralized power generating options are achieved by their installation on rooftops, integration into building facades, or placement as ground-mounted arrays.

B. Boost Converter

A boost converter is a DC–DC converter that raises the voltage (1:1.5) between its source and load, or input and output. When the desired voltage level is higher than the input voltage that is available, it is commonly employed in many different applications. A crucial part of many power supply and energy management systems, the boost converter works on the idea of energy storage and release. Several appropriate control techniques may be used for the creation of a constant voltage. An energy discharge phase and an energy storage phase are used in the converters to transfer energy. When the switch is in mode 1, the supply voltage, Vpv, causes the inductor, L, to begin storing energy as a magnetic field. The inductor current rises as a consequence.

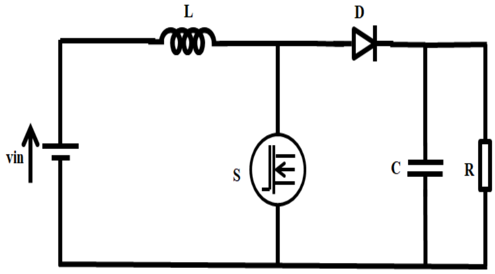

Fig. 3. Boost Converter

The output circuit is separated from the source side and no current passes through the diode and output side because diode D is reverse biased.

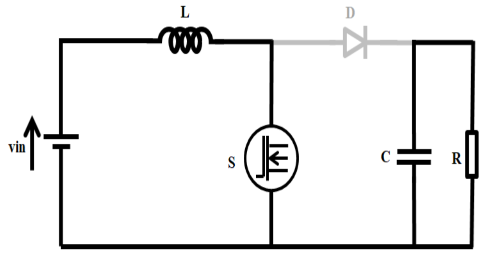

Fig. 3. (a) Mode I

Current flows across the inductance, switch, and source when the switch is closed, acting in the short circuited mode. Boost converters are basic converters that get their name from what they do. The boost converter is made up of the following components: load resistor R, filter capacitor C, diode D, and power switch S.

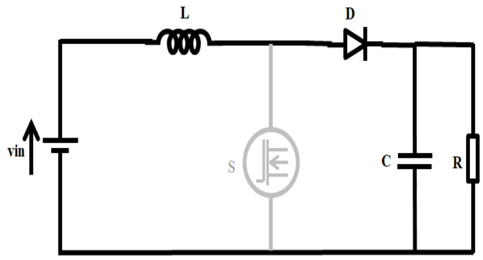

Fig. 3. (b) Mode II

In mode 2, the switch is in the OFF condition, the inductor reverses its polarity and acts as a source. Inductor does not allow sudden changes in the current according to Lenz law but it will allow smooth amount of current to flow through diode. The capacitor is getting charged from the source and the inductance present.

For the inductor L, the voltage-current relationship is:

$$i = \frac{1}{L}\int_0^t V dt + i_0 \tag{3}$$

Or

$$V = L\frac{di}{dt} \tag{4}$$

For a rectangular pulse that is constant:

$$i = \frac{Vt}{L} + i_0 \tag{5}$$

Upon switching the transistor M_1:

$$i_{pk} = \frac{(V_{in} - V_{Trans})T_{on}}{L} + i_0 \tag{6}$$

$$\Delta i = \frac{(V_{in} - V_{Trans})T_{on}}{L} \tag{7}$$

And the current is as follows when the transistor is turned off:

$$i_0 = i_{pk} - \frac{(V_{out} - V_{in} + V_D)T_{off}}{L} \tag{8}$$

$$\Delta i = \frac{(V_{out} - V_{in} + V_D)T_{off}}{L} \tag{9}$$

In this case, V_{Trans} is the voltage drop across transistor M_1, and V_D is the voltage drop across diode D_m.

Find V_{out} by equating via delta i:

$$\frac{(V_{in} - V_{Trans})T_{on}}{L} = \frac{(V_{out} - V_{in} + V_D)T_{off}}{L} \tag{10}$$

$$V_{in} - V_{Trans}D = (V_{out} + V_D)(1 - D) \tag{11}$$

$$V_{out} = \frac{V_{in} - V_{Trans}D}{(1-D)} - V_D \tag{12}$$

Ignoring the voltage dips across the transistor and the diode

$$V_{out} = \frac{V_{in}}{1-D} \tag{13}$$

The total of the stored energy from the source side and the inductor will equal the voltage across the capacitor. There is a difference between the input and output voltages. As a result, the output DC voltage is increased.

C. DFIG

One kind of wind turbine generator that is frequently used in contemporary wind power systems is the DFIG.

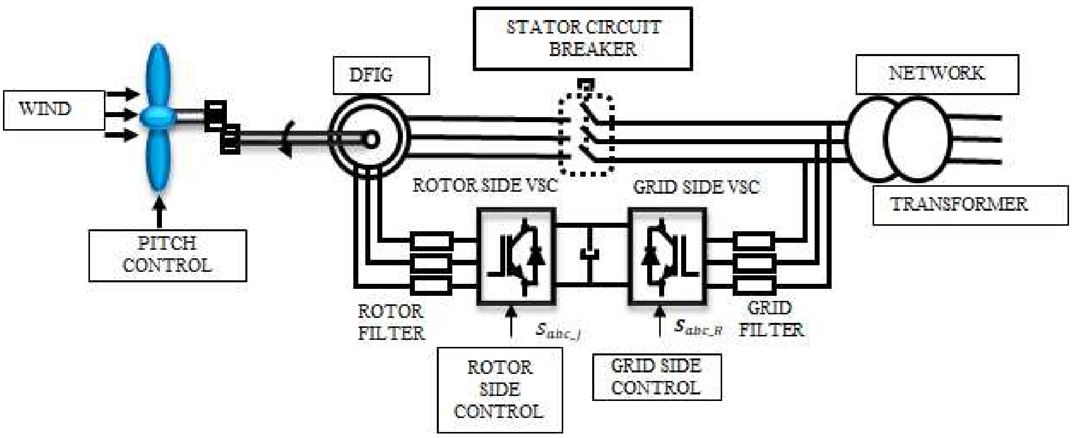

Fig. 4. DFIG

It comprises of a power electronic converter that connects the wound rotor induction generator to the grid. DFIGs, as opposed to fixed-speed generators, provide variable-speed operation, which improves wind energy consumption at all wind speeds. A partial-scale power converter on the rotor side allows DFIGs to change the rotor speed independently of the grid frequency, which is one of its primary features. This function improves grid stability and energy collection efficiency. The rotor-side converter regulates the rotor current during regular operation, enabling the generator to produce the most power by modifying the torque and speed.

$$\vec{V}^S{}_S = R_S \vec{i}^S{}_S + \frac{d\vec{\lambda}^S{}_S}{dt} \tag{14}$$

$$\vec{V}^S{}_r = R_r \vec{i}^S{}_r + \frac{d\vec{\lambda}^S{}_r}{dt} - j\omega_m \vec{\lambda}^S{}_r \tag{15}$$

$$\vec{\lambda}^S{}_S = L_S \vec{i}^S{}_S + L_m \vec{i}^S{}_r \tag{16}$$

$$\vec{\lambda}^S{}_r = L_m \vec{i}^S{}_S + L_r \vec{i}^S{}_r \tag{17}$$

The magnetizing inductance is called Lm, the angular frequency is m, the resistance is R, and the inductance is L. Voltage is represented by v. I stand for the present. The rotor and stator quantities are denoted by the subscripts r and s, respectively.

D. Three Phase VSI

A power electronic device utilized in renewable energy systems and AC motor drives is a three-phase VSI. DC electricity is transformed into three-phase AC power via it. It is made up of six power switches, usually three pairs of insulated-gate bipolar transistors (IGBTs). Through a precise series of switchbacks, it generates three-phase AC electricity from a DC input.

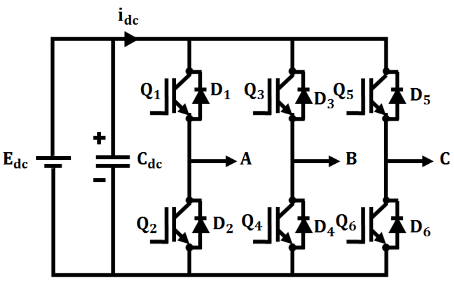

Fig. 5. Three Phase VSI

VSIs are appropriate for a variety of applications needing variable-speed motor control because they provide fine control over the output frequency, amplitude, and phase angle. They support grid stability and energy efficiency in a variety of sectors through their widespread usage in adjustable-speed drives, uninterruptible power supply (UPS), and grid-connected renewable energy systems.

IV. Results And Discussion

TABLE I. Parameter Specifications

Parameters	Rating
DFIG	
Number of turbine	1
Voltage	575 Volts
Power	10 KW
Boost Converter	
V_{in}	$6V$
L	$3mH$
C	$810\mu F$
R	6Ω

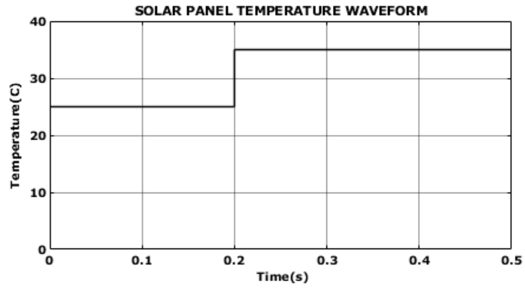

Fig. 6. Temperature waveform

Fig. 6 shows that the temperature waveform is 25°C at 0.2s. After 0.2s, the waveform rises to 35°C and then stays constant.

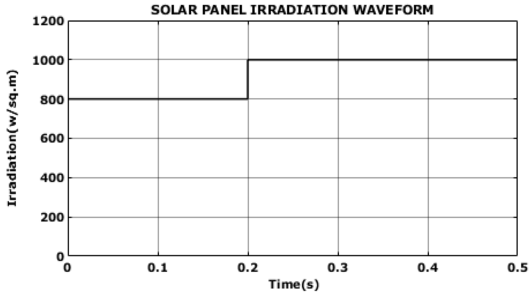

Fig. 7. Irradiation Waveform

Fig. 7 illustrates the irradiation waveform at 0.2 seconds, which is 800 W/Sq.m. After 0.2 seconds, the waveform increases to 1000W/Sq.m and then stabilizes.

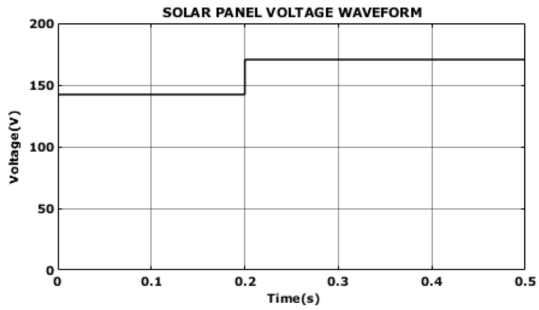

Fig. 8. Voltage waveform

The voltage waveform is 140V at 0.2 seconds, according to Fig. 8. The waveform rises to 175V after 0.2s and stays there.

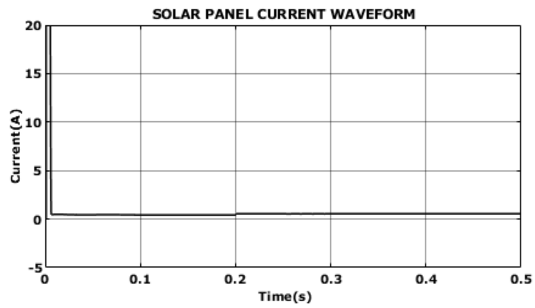

Fig. 9. Current waveform

Fig. 9 shows that the current waveform grows during a 0.02 second period, from 0 A to 20 A. The waveform drops to 1A after 0.02s and stays there.

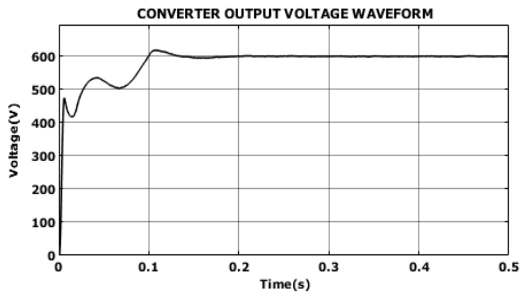

Fig. 10. Converter output voltage waveform

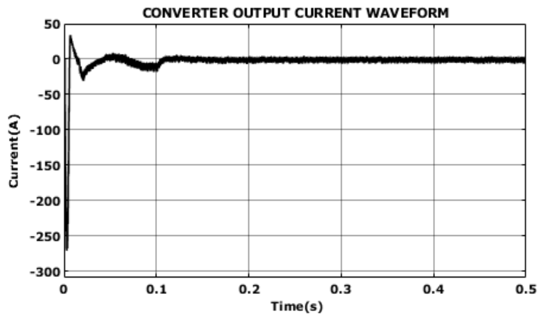

Fig. 11. Current waveform

The waveform in Fig. 10 grows from 0 V to 600 V over 0.1 seconds while staying constant.

The waveform grows from -250A to 40A during a 0.02s period, as seen in Fig. 11. The waveform drops to 0A after 0.02s and stays steady after that.

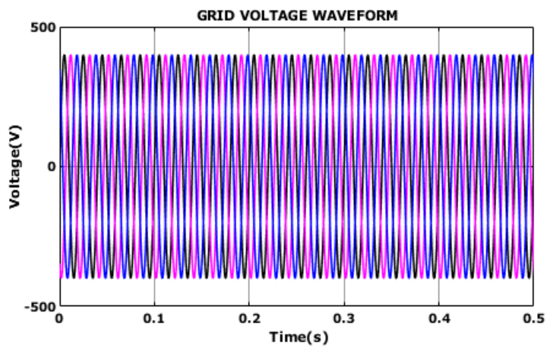

Fig. 12. Grid voltage waveform

The grid voltage waveform is shown in Fig. 12 as being constant between -400V and 400V.

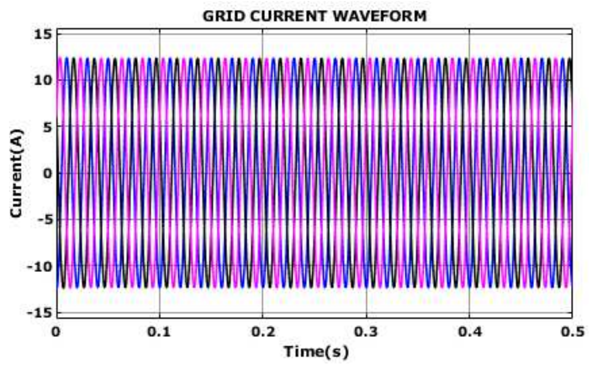

Fig. 13. Grid current waveform

The grid current waveform is shown in Fig. 13 as being constant between -12A and 12A.

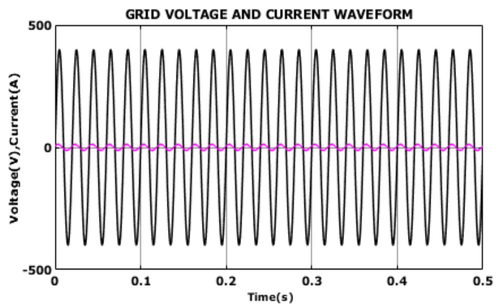

Fig. 14. Grid voltage and current waveform

The waveform is shown in Fig. 14 as being constant between -400V and 400V at 0A.

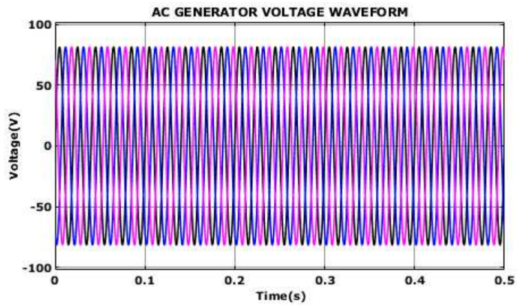

Fig. 15. AC Generator voltage waveform

The voltage waveform of the AC generator is constant between -75V and 75V, as shown in Fig. 15.

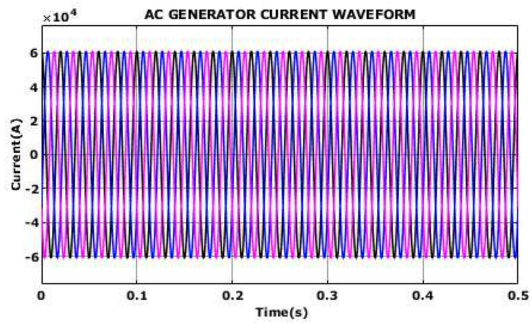

Fig. 16. AC Generator current waveform

The AC generator current waveform is seen in Fig. 16 to be constant between -6A and 6A.

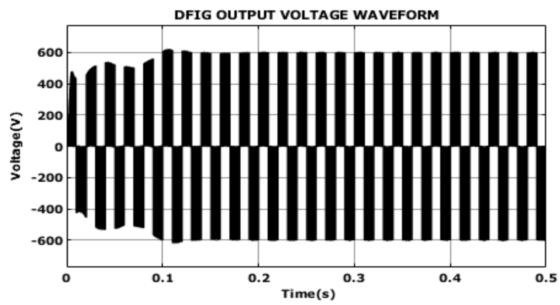

Fig. 17. DFIG output voltage waveform

The waveform of the DFIG is constant between -600V and 600V, as seen in Fig. 17.

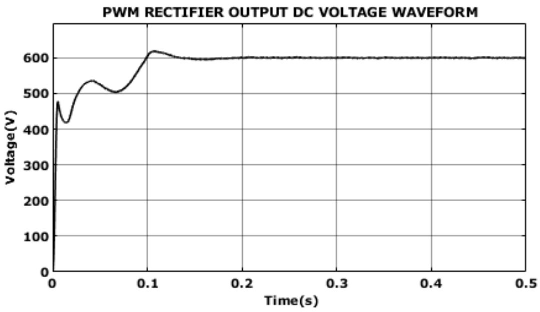

Fig. 18. PWM Rectifier output DC voltage waveform

Fig. 18 shows the DC voltage waveform of the PWM rectifier output as it grows from 0V to 600V during a 0.2-second period and stays constant.

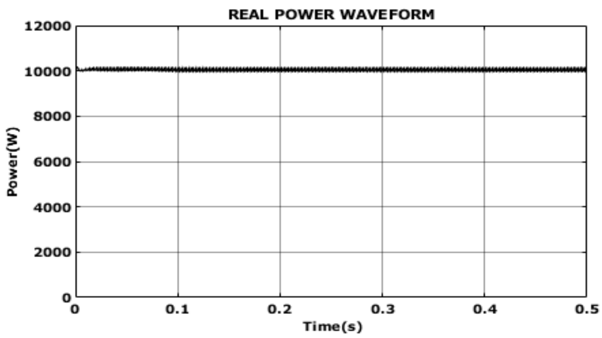

Fig. 19. Real power waveform

The real power waveform at 10,000W is constant, as seen in Fig. 19.

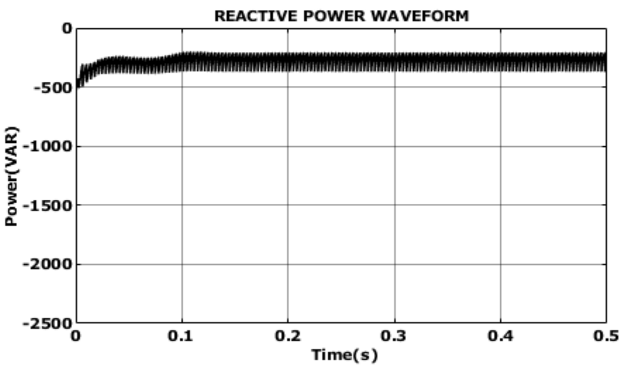

Fig. 20. Reactive Power Waveform

Reactive power waveform at -300VAR is constant, as seen in Fig. 20.

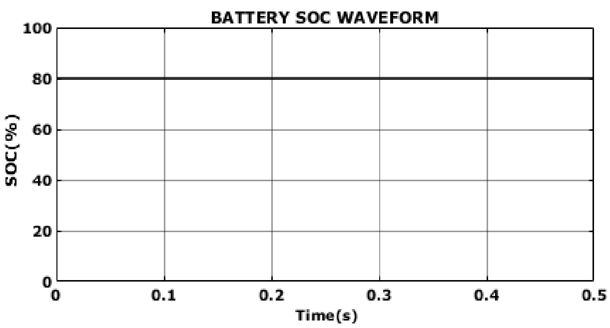

Fig. 21. Battery SOC Waveform

The battery SOC waveform is steady at 80%, as seen in Fig. 21.

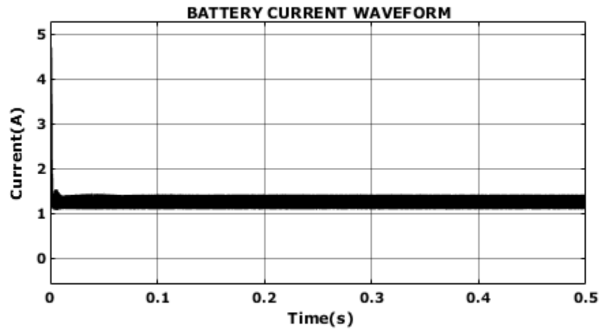

Fig. 22. Battery Current Waveform

The battery current waveform is shown in Fig. 22 as being constant at 1A.

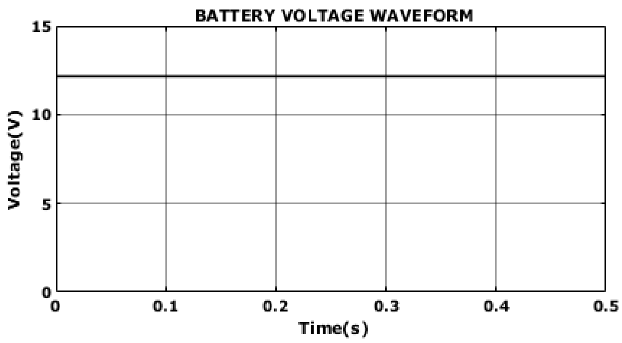

Fig. 23. Battery Voltage waveform

The battery voltage waveform is seen in Fig. 23 to be constant at 12 V.

V. Conclusion

This study offered a revolutionary hybrid energy-source-based method for grid-tied electric car charging. It enables communication from the car to the grid and from the grid to the vehicle. Through a boost converter, electricity from the solar system is sent to the inverter. For the purpose of controlling the DC bus voltage and boost converter current, an energy management plan is recommended. The WECS's DFIG and AC-DC conversion are made easier by a PWM rectifier that is managed by a PI controller. Through a three phase VSI, the converter's output which transforms the DC voltage into an AC voltage is delivered into the grid. A sinusoidal waveform is produced by reducing harmonics using the LC filter. Any energy left over is used to power the linear and non-linear loads in the next home. In this case, the electric car battery is charged by the PV panel, wind, and AC generator. The bidirectional power flow makes it easier to recycle excess electricity back into the grid for other purposes at times when there is less demand for EVs and to charge EV batteries effectively. Battery charging efficiency has been effectively increased via the creation and consideration of hybrid charging systems.

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